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Studierichting: Informatica
Oefeningenreeks 2D nr. 4.1

$f(x) = \lfloor x \rfloor - 4$

$\lim_{x \to 0-} (\lfloor x \rfloor - 4) = -1 - 4 = -5$

$\lim_{x \to 0+} (\lfloor x \rfloor - 4) = 0 - 4 = -4$

Limiet bestaat niet.

$\lim_{x \to 4-} (\lfloor x \rfloor - 4) = 3 - 4 = -1$

$\lim_{x \to 4+} (\lfloor x \rfloor - 4) = 4 - 4 = 0$

Limiet bestaat niet.

$\lim_{x \to \frac{1}{2}} (\lfloor x \rfloor - 4) = 0 - 4 = -4$

Limiet: -4

$\lim_{x \to \sqrt{2}} (\lfloor x \rfloor - 4) = 1 - 4 = -3$

Limiet: -3

$\lim_{x \to -\frac{5}{4}} (\lfloor x \rfloor - 4) = -2 - 4 = -6$

Limiet: -6

$\lim_{x \to +\infty} (\lfloor x \rfloor - 4) \geq +\infty - 5 = +\infty$

Limiet: $+\infty$

Want: $x - 5 < \lfloor x \rfloor - 4 \leq x - 4$

References